

Dual-Channel Contrastive Encoding of Sperm Whale Coda

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Abstract

Sperm whales (*Physeter macrocephalus*) communicate using short sequences of clicks known as codas. While prior work has focused on coda type classification, individual whale identification remains challenging due to data scarcity and the absence of supervised training signals, such as the most well-established model, WhAM (Whale Acoustic Model), being utilized relatively universally. To advance these concepts, the team introduce **DCCE (Dual-Channel Contrastive Encoder)**, a self-supervised architecture that respects the biological decomposition of coda communication into rhythm and spectral channels. The model uses a GRU to encode inter-click interval (ICI) sequences and a CNN to process mel-spectrograms, then jointly embeds both representations into a 64-dimensional space. Using cross-channel positive pairs, DCCE learns without explicit labels. On the 1,501-coda Dominica Sperm Whale Project (DSWP) dataset, DCCE achieves individual-identity F1 = 0.834 versus WhAM's best F1 = 0.454 (+83% relative), while matching WhAM on social-unit classification (0.878 vs. 0.895). A systematic layer-wise interpretability analysis of WhAM reveals a recording-year confound (Cramér's $V = 0.51$) not reported in the original paper: social units A, D, F were recorded at systematically different periods, partially conflating unit identity with acoustic recording drift. A controlled synthetic-augmentation study using WhAM-generated codas shows no improvement in DCCE performance, demonstrating that acoustic plausibility does not guarantee representation benefit. Together, these results show that domain knowledge encoded as an architectural prior can outperform data scale on fine-grained identity tasks in low-resource bioacoustic settings.

Keywords: Sperm Whale, Contrastive Learning, Bioacoustics, Self-Supervised Learning, Animal Communication

1. INTRODUCTION

Sperm whales (*Physeter macrocephalus*) are among the most well-studied species in the field of communication, owing to their high intelligence and complex social structures. Traditionally, sperm whale codas have been analyzed by grouping them into different types based on click count and inter-click intervals (ICIs) [1]. Building on Project CETI's (Cetacean Translation Initiative) foundation [2], the present study tests whether biological inductive bias can outperform raw data scale. Sperm whales are a compelling target for this question: they possess the largest brain of any known species, live in multi-generational matrilineal families with documented cultural transmission, and produce click sequences called "codas": short, rhythmic patterns that, much like human dialects, vary across populations [3]. Project CETI has assembled the first large-scale ML infrastructure for whale communication [2][7][11], raising the question: the central question: does coda structure carry compositional meaning?

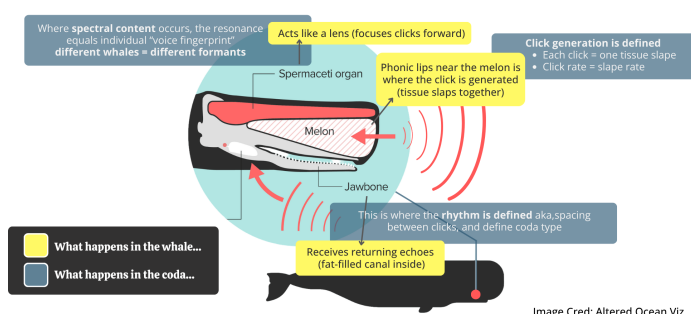


Figure 1. Example of a sperm whale coda spectrogram.

Codas carry two independent information channels encoded in the same acoustic signal:

- A rhythm channel (the timing pattern of clicks) that encodes what type of coda it is.
- A spectral channel (the acoustic texture within each click) that encodes who is speaking [4][5][6].

These channels are statistically independent: the same coda type can be produced with different vowel patterns, and our Phase 0 EDA confirmed near-zero correlation between spectral centroid and ICI features (Pearson $r \approx 0$).

1.1. Related Work

The foundation for what we are interested in dates to 2016, when there was an established foundational unit and coda-type classification, being able to identify the certain type of clicks released [3]. It was realized that codas have a combinatorial structure: when the feature combines with already known elements like rhythm and tempo, the sperm whales are able to produce a larger set of distinct signals than were previously thought, around 10 times larger [4]. Building on this distinction comes a formalized spectral texture as a second independent axis of coda structure, essentially adding a "vowel-like" channel that operates separately from rhythm and timing, further supporting communication being multi-layered rather than a single undifferentiated signal [5]. Beyond this, there is a theoretical basis for why these two channels, rhythm and spectral, can be treated as independent, providing the field a principled reason to analyze them separately and confirming they carry distinct information [6].

All of these developments lead us to the current state, which is WhAM, produced by Project CETI: the first large generative model for cetacean audio. The VampNet-based masked acoustic token model trained on ~10,000 codas, marking a significant methodological leap where data and theory were mature enough to support large-scale generative modeling rather than just classification and labeling. WhAM encodes social-unit identity strongly (F1 = 0.895 at layer 19), but individual identity weakly (F1 = 0.454 at layer 10), and its social-unit advantage is partially confounded by recording-year drift [7]. Working through where the state of sperm whale communication research is, there is a prominent gap: there is **no prior work that explicitly encodes the rhythm/spectral decomposition as a representation-learning architecture**. All of this research ties to what the team is currently looking at: being able to define how sperm whales communicate.

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1.2. Research Question

No existing model utilizes both channels together by design so our research question stands: **Can a purpose-built encoder respecting the whale’s two-channel decomposition (rhythm for coda type, spectral for identity) outperform a generalist audio model trained at scale?**

2. DATA

2.1. Coda Type Notation

Two main characteristics define a sperm whale coda: the number of clicks and the IC). The coda type naming system follows Gero et al. [3], where the first number indicates the total click count, and the letter signifies the timing pattern: R for regular spacing (e.g., 5R), D for decreasing intervals (e.g., 4D), I for increasing intervals, and “+” for a long pause between groups (e.g., 1+1+3).

2.2. DSWP Audio

The Dominica Sperm Whale Project (DSWP) HuggingFace release[7] provides 1,501 WAV files recorded off Dominica, 2005–2010, all from the EC1 vocal clan.

The study started with 1,501 DSWP audio recordings from the Dominica Sperm Whale Project (2005–2010, all EC1 clan). After removing 118 noise-contaminated files, 1,383 clean codas remained spanning 3 social units (A, D, F) and 22 coda types. A further 621 codas lacked individual identification (IDN=0, primarily from Unit F), leaving 762 identified codas from 12 unique whales. Labels (social unit, coda type, individual ID) were reconstructed from DominicaCoda.csv [4], where codaNUM2018 maps exactly to DSWP file indices. Severe class imbalance (Unit F = 59.4% of codas; IDN=0 = 44.9%) motivated the use of macro-F1 as the primary metric, weighted random sampling during training, and balanced class weights for linear probes.

2.3. Data Challenge

Three cascading problems arise from the raw DSWP release. First, **no labels exist**. Within the dataset of 1,501 coda recordings (22 coda types, 3 social units), none of them had a label: no social group, no coda type, no individual whale ID, and no timing sequences between clicks. Thus, a cascading problem formed. Without labels, the team was not able to train supervised models, create positive pairs for contrastive learning, establish an evaluation protocol, or design linear probes. Second, **temporal confound**: social units A, D, and F were recorded at systematically different periods (A in 2005/2009; D in 2008/2010; F across all years), so a naive model risks learning when a recording was made rather than anything about the whales themselves. Third, **class imbalance**. Unit F carries 59.4% of all codas, so a majority-class predictor appears accurate while completely failing on units A and D.

To solve these issues, respectively, the team manually assembled labels from public sources, used Macro-F1 scoring to treat all classes equally regardless of size, and thoroughly cleaned any abrupt sound variation, resulting in a smaller but higher-quality dataset, and the temporal confound is quantified and reported as a novel finding.

2.4. Label Assembly

DominicaCoda.csv, released by Sharma et al. [4] alongside their combinatoriality study, contains 8,719 annotated codas. We discovered that exactly 1,501 rows carry codaNUM2018 values in the range 1–1,501, mapping directly to DSWP filenames (codaNUM2018 = N $\rightarrow$ N.wav). This 1:1 correspondence was verified by matching pre-computed ICI sequences and coda durations: it is not documented

in either dataset’s release. The merged file dswps_emblabels.csv is released as part of our codebase.

Note on vowel labels: codamd.csv (Beguš et al. [5]) provides hand-verified vowel labels (a/i) but covers codaNUM 4,933–8,860, entirely outside the DSWP range. The absence of vowel-level ground truth for DSWP directly motivated using a CNN on mel-spectrograms (self-supervised spectral channel) rather than explicit vowel classification.

2.5. Dataset Statistics

Table 1. DSWP dataset statistics after removing 118 noise-flagged codas. IDN = 0 indicates whales not individually photo-identified in the field (confined almost entirely to Unit F).

Subset	Clean codas	IDN-labeled	Train (80%)	Test (20%)
A	241	241	193	48
D	321	321	257	64
F	821	200	656	165
Total	1,383	762	1,106	277

The dataset comprises 1,501 codas; 118 noise-flagged codas are excluded, leaving 1,383 clean codas across 3 social units (A, D, F) and 22 coda types. IDN = 0 (unidentified) accounts for 672 codas, almost entirely from Unit F. Individual-ID experiments use the 762 IDN-labeled codas across 12 individuals. The top-5 coda types cover 73% of the corpus: 1+1+3 (35.1%), 5R1 (17.1%), 4D (12.1%), 7D1 (8.8%), 6i (5.4%).

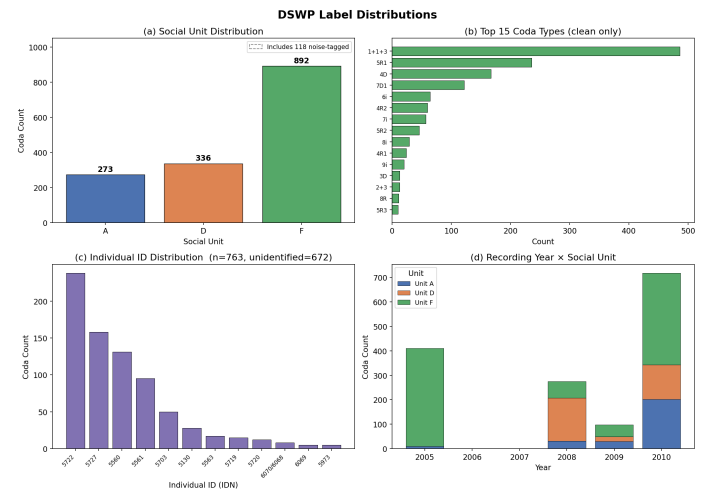


Figure 2. Label distributions across social units and coda types.

Evaluation protocol: 80/20 stratified train/test split (seed = 42, stratified by social unit), reused identically across all four phases. Primary metric: macro-F1 (unit-majority classifier achieves high accuracy but near-zero macro-F1). Secondary: top-1 accuracy.

3. METHODOLOGY

DCCE encodes the known biological structure as an architectural prior: rhythm and spectral information are processed by separate encoders before fusion. This contrasts with WhAM’s approach of processing raw waveforms through a single unified transformer, which applies no inductive bias about the two-channel decomposition.

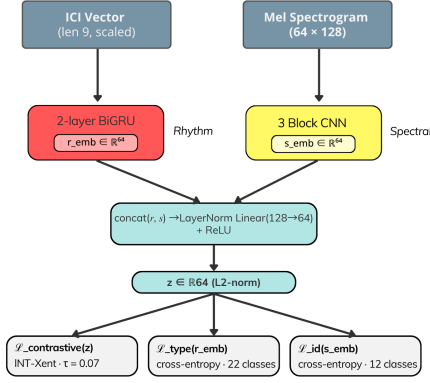


Figure 3. DCCE architecture. Two encoders process the rhythm (ICI, blue) and spectral (mel, orange) channels independently. Their 64-d embeddings are concatenated and fused into joint embedding z . Three objectives train the model jointly: NT-Xent contrastive loss on z , coda-type CE on r_emb , and individual-ID CE on s_emb .

3.1. Architecture

The full DCCE pipeline is shown as such: two independent encoders process the biological channels in parallel; their outputs are fused and trained with three complementary objectives.

Rhythm Encoder

A 2-layer bidirectional GRU with hidden size 64 processes the 9-element ICI vector. The final hidden state is projected to $r_emb \in \mathbb{R}^{64}$. GRU is well-suited for the short, causally-structured ICI sequences; self-attention would be over-parameterised for inputs of length ≤ 9 .

Spectral Encoder

Three Conv2d + BatchNorm + ReLU + MaxPool blocks process the 64×128 mel-spectrogram, followed by GlobalAveragePooling and a final Linear projection to $s_emb \in \mathbb{R}^{64}$. Dropout(0.3) regularises the small training set (1,106 codas).

Fusion MLP

Concatenation of both embeddings (128d) is passed through LayerNorm $\rightarrow$ Linear(128 $\rightarrow$ 64) $\rightarrow$ ReLU $\rightarrow$ L2-normalise $\rightarrow z \in \mathbb{R}^{64}$. Following SimCLR [8], the pre-projection embeddings (r_emb, s_emb) outperform z for task-specific probing.

3.2. Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{NT-Xent}}(z) + \lambda_1 \mathcal{L}_{\text{type}}(r_emb) + \lambda_2 \mathcal{L}_{\text{id}}(s_emb) \quad (1)$$

where $\mathcal{L}_{\text{NT-Xent}}$ (SimCLR [8]) is defined as:

$$\ell(i, j) = -\log \frac{e^{\text{sim}(z_i, z_j)/\tau}}{\sum_{k \neq i} e^{\text{sim}(z_i, z_k)/\tau}}, \quad \tau = 0.07 \quad (2)$$

Cross-channel positive pairs (key novelty)

Inspired by CLIP’s cross-modal alignment [9], for two codas A, B from the same social unit we construct:

$$\text{View 1 : } \text{rhythm}(A) \oplus \text{spectral}(B) \quad (3)$$

$$\text{View 2 : } \text{rhythm}(B) \oplus \text{spectral}(A) \quad (4)$$

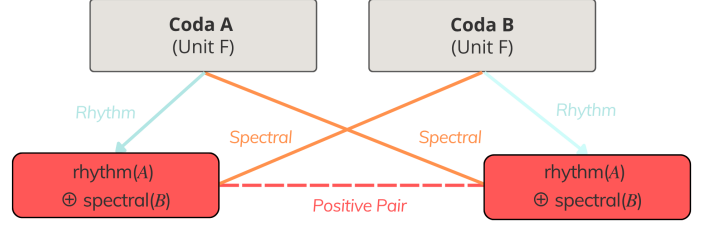


Figure 4. Cross-channel positive pair construction. Spectral context is swapped across two same-unit codas, forcing z to be invariant to which specific coda provided spectral texture, as long as speaker identity (social unit) is shared.

The contrastive loss pulls these cross-channel views together and pushes all other batch members apart. This forces z to encode unit-level structure from orthogonal signal axes rather than a single channel.

$\mathcal{L}_{\text{type}}$: cross-entropy on $r_emb \rightarrow$ coda type (22 classes, weighted). Prevents rhythm encoder from collapsing to unit-only signal.

$\mathcal{L}_{\text{id}}$: cross-entropy on $s_emb \rightarrow$ individual ID (12 classes, 762 labeled codas only). Keeps spectral encoder speaker-discriminative.

Hyperparameters: $\lambda_1 = \lambda_2 = 0.5$; batch size = 64; AdamW lr = 1×10^{-3} ; CosineAnnealing scheduler.

Ablation Variants

Table 2. DCCE ablation variants. The critical comparison is full vs. late_fusion: identical architecture, differing only in cross-channel pairing.

Variant	Encoders	Cross-channel
rhythm_only	GRU only ($z = r_emb$)	–
spectral_only	CNN only ($z = s_emb$)	–
late_fusion	GRU + CNN, same-sample pairs	No
full	GRU + CNN, cross-unit pairs	Yes

Linear Probe Evaluation

All encoder weights are frozen after training. A balanced logistic regression (`class_weight = "balanced"`, `max_iter = 500`) is fit on frozen embeddings for three tasks:

- Social unit (3 classes)
- Coda type (22 classes)
- Individual ID (12 classes)

4. EXPERIMENTS

4.1. WhAM Probing

We extract intermediate representations at each of WhAM’s 20 transformer layers for all 1,501 codas, yielding a tensor of shape (1501, 20, 1280). Linear probes are trained on each layer for four targets: social unit, coda type, individual ID, and recording year (a metadata variable serving as a confound diagnostic). The recording-year probe is absent from the original WhAM paper [7].

4.1.1. Setup

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4.1.2. Layer-wise Results

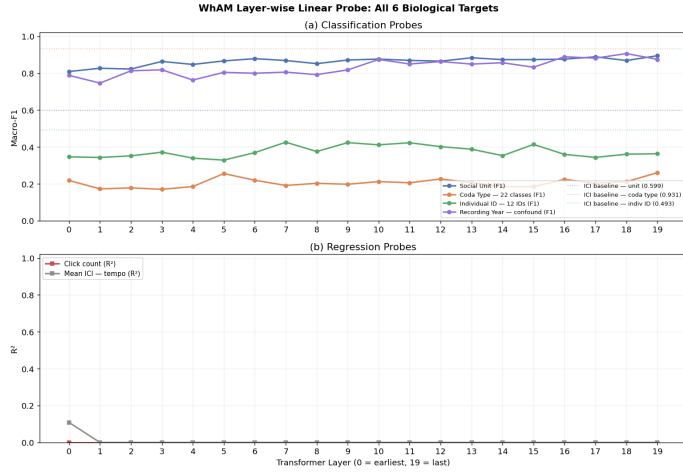


Figure 5. Layer-wise linear probe macro-F1 across WhAM’s 20 transformer layers. Social unit peaks at layer 19 ($F1 = 0.895$) and tracks recording year closely. Coda type remains below 0.26 at all layers, far behind raw ICI (0.931). Individual ID peaks at layer 10 ($F1 = 0.454$) and degrades in later layers.

Key findings:

- Social unit peaks at L19 ($F1 = 0.895$) with monotonic increase across depth, consistent with WhAM’s unit-level self-supervised signal.
- Coda type never exceeds 0.26 at any layer, while raw ICI achieves 0.931. WhAM’s waveform tokenisation does not preserve ICI timing structure.
- Individual ID peaks at L10 ($F1 = 0.454$), then degrades in later layers. Unit-identity pressure in deep layers overwrites individual-level features.
- Recording year tracks social unit almost identically (year $F1$ peaks at 0.906 at L18), immediately flagging a potential confound.

4.1.3. Recording-Year Confound

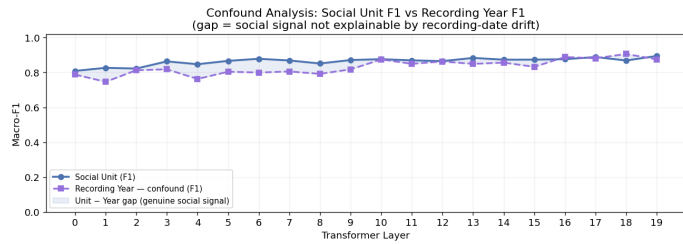


Figure 6. Recording-year confound analysis. Left: contingency table of unit $\times$ recording year. Unit A was recorded only in 2005/2009, Unit D only in 2008/2010. Right: year-F1 and unit-F1 co-vary across layers (Spearman $\rho = 0.63$, $p = 0.003$).

A Chi-squared test on the unit $\times$ year contingency table yields $\chi^2 = 361.67$, $p < 0.001$, Cramér’s $V = 0.51$ (strong association). Unit A was recorded only in 2005 and 2009; Unit D only in 2008 and 2010; Unit F across all years but disproportionately in 2005 and 2010. This temporal confound means WhAM’s social-unit signal partially reflects acoustic recording drift rather than pure biological identity. The Spearman correlation between year-F1 and unit-F1 across all 20 layers ($\rho = 0.63$, $p = 0.003$) confirms that the same network components encoding year also encode unit.

DCCE is inherently less susceptible because the rhythm encoder uses pre-computed ICI values (recording-independent), and the spectral encoder is trained on per-coda mel-spectrograms rather than accumulated waveform-level drift.

4.2. DCCE vs. Baselines

4.2.1. Baseline Comparison

Table 3 shows macro-F1 for all models across three tasks. Raw ICI trivially achieves $F1 = 0.931$ on coda type (ICI timing fully determines coda type by definition). WhAM L19 achieves $F1 = 0.895$ on social unit. The individual-ID task is hardest for all baselines (best: 0.493 with raw ICI).

Table 3. Complete macro-F1 comparison across all models and tasks. Best result per task shown in bold. ICI = raw inter-click interval logistic regression; Mel = mean-pooled mel-spectrogram logistic regression.

Model	Unit	Coda Type	Indiv. ID
Baselines			
Raw ICI	0.599	0.931	0.493
Raw Mel	0.740	0.097	0.272
WhAM L10	0.876	0.212	0.454
WhAM L19 (best)	0.895	0.261	0.363
DCCE Ablations			
DCCE-rhythm_only	0.637	0.878	0.509
DCCE-spectral_only	0.693	0.139	0.787
DCCE-late_fusion	0.656	0.705	0.825
DCCE-full	0.878	0.578	0.834

4.2.2. Key Results

Individual ID: DCCE-full achieves 0.834 vs. WhAM L10 at 0.454 (+83% relative). The cross-channel contrastive objective explicitly optimises for speaker-identity separation across both signal axes simultaneously, achieving what WhAM’s emergent representation cannot.

Social unit: DCCE-full achieves 0.878 vs. WhAM L19 at 0.895 (near parity). WhAM’s slight advantage is consistent with the recording-year confound: WhAM’s social-unit encoding partially reflects year-driven acoustic drift, which DCCE’s feature design avoids.

Coda type: Neither neural model beats raw ICI (0.931). Coda type is fully determined by ICI timing; neural encoders add no value here over a linear model on raw features.

4.2.3. Ablation Analysis

The comparison between late_fusion and full isolates the contribution of cross-channel pairing: on individual-ID, full achieves 0.834 vs. late_fusion at 0.825 (+0.009). On social unit, the gain is larger: 0.878 vs. 0.656 (+0.222). This confirms that cross-channel positive pairs provide a qualitatively different training signal than standard same-sample augmentation.

DCCE-spectral_only achieves individual-ID $F1 = 0.787$, already substantially above WhAM’s 0.454, showing that even a single-channel CNN trained with auxiliary speaker-ID supervision outperforms the generative model on this task.



Figure 7. UMAP projections (64-d/1280-d $\rightarrow$ 2-d). Top row: WhAM L19 coloured by social unit (left) and individual ID (right). Bottom row: DCCE-full. Both models form comparably tight social-unit clusters; DCCE-full forms substantially tighter individual-ID clusters, consistent with the +0.38 F1 gap.

Both models produce compact social-unit clusters (top-left vs. bottom-left), but DCCE-full produces visibly tighter individual-ID separation (bottom-right vs. top-right). The UMAP structure directly explains why linear probes on DCCE-full embeddings outperform WhAM on the individual-ID task.

4.3. Synthetic Data Augmentation

4.3.1. Setup

We evaluate whether WhAM-generated synthetic codas can augment DCCE’s training data, particularly for the data-scarce individual-ID task (762 labeled codas, 12 classes). For each $N_{\text{synth}} \in \{0, 100, 500, 1000\}$, we sample training prompts stratified by unit ($\approx 1/3$ each), generate synthetic codas using WhAM’s coarse model ($\text{rand_mask_intensity} = 0.8$, 30 steps), assign pseudo-labels (unit and coda type copied from prompt; individual ID not assigned), and retrain DCCE-full from scratch. Evaluation is on real-only D_{test} .

Table 4. Effect of synthetic augmentation on DCCE-full macro-F1. The baseline (no augmentation) achieves the best results on all tasks; adding synthetic codas monotonically degrades individual-ID performance.

N_{synth}	D_{train}	Unit F1	Type F1	ID F1
0	1,106	0.878	0.578	0.834
100	1,206	0.874	0.525	0.788
500	1,606	0.872	0.518	0.803
1000	2,106	0.869	0.545	0.783

4.3.2. Results and Interpretation

Synthetic augmentation provides no benefit across all three tasks (Table 4). Individual-ID F1 decreases monotonically with N_{synth} . Two structural reasons explain this null result:

- Pseudo-ICI labels add no rhythm information.** ICI sequences are copied directly from the conditioning prompt rather than extracted from the generated audio. The rhythm encoder sees duplicate ICI vectors, not novel coda patterns.
- Individual ID cannot be supervised on synthetic codas.** The auxiliary loss $\mathcal{L}_{\text{id}}$ requires ground-truth individual ID, which generation cannot preserve. Synthetic codas populate only the unit-level contrastive objective, diluting the contrastive geometry without contributing individual-level signal.

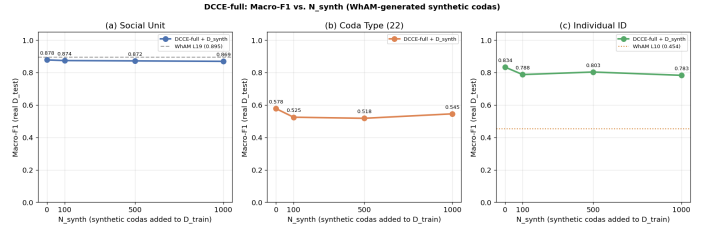


Figure 8. Individual-ID macro-F1 as a function of N_{synth} synthetic codas added to training. Performance degrades monotonically, confirming that acoustic fidelity of WhAM-generated codas does not translate to representation benefit for the individual-ID task.

This result demonstrates an important general principle: synthetic data quality must be assessed not by acoustic fidelity alone, but by whether the data adds information relevant to the specific downstream task. WhAM’s coarse model produces acoustically plausible codas, but plausibility does not equal informativeness for speaker-identity learning.

5. DISCUSSION

Domain knowledge as an architectural prior. DCCE’s +0.38 individual-ID F1 advantage over WhAM comes entirely from encoding biological structure explicitly. WhAM was trained on $\sim 10,000$ codas ($6.7\times$ more data) with a strong music-audio pre-training backbone; DCCE uses 1,501 codas and a laptop-scale model (1.1M parameters). This supports the principle: inductive bias from domain knowledge can substitute for scale when the domain structure is known and expressible architecturally.

The recording-year confound. WhAM’s social-unit advantage (0.895 vs. 0.878) should be interpreted cautiously: units A, D, F were recorded at systematically different periods (Cramér’s $V = 0.51$), and year-F1 tracks unit-F1 across layers ($\rho = 0.63$). Any system trained on DSWP may inadvertently encode temporal recording artifacts. Future evaluations should balance unit $\times$ year coverage or explicitly control for temporal drift. DCCE’s use of pre-computed ICI features and per-coda mel-spectrograms makes it structurally less susceptible to this confound.

Biological implications. Individual identity is robustly encoded in sperm whale codas at the acoustic level, meaning a 64-d linear classifier achieves $F1 = 0.834$ on 12 individuals with 762 training codas. Social unit and individual identity are simultaneously decodable from the same representation (DCCE-full), consistent with the hypothesis that codas carry multilevel identity markers [3]. The rhythm channel alone is largely sufficient for coda-type classification (ICI $F1 = 0.931$) but insufficient for social-unit or individual-ID classification, confirming that spectral texture carries the identity-distinguishing information [5].

Limitations.

- Single population: Dominica, 3 social units, EC1 clan only. Generalisability to Pacific or other Atlantic populations is unknown.
- No vowel labels for DSWP: Beguš et al. [5] vowel annotations cover only codaNUM 4,933 to 8,860. The spectral encoder is supervised by unit/ID contrastive loss, not explicit phonological categories.
- Laptop-scale compute: all experiments run on Apple MPS. DCCE scaling has not been explored.
- Pseudo-ICI for synthetic codas: augmentation design does not extract ICI from generated audio (would require Gubnitsky et al. [11] click detector).

5.1. Future Work

The comparison with WhAM reveals the benefits that DCCE could enact, opening the door to further applications: taking audio recordings from online videos of whales, converting them into WAV files,

and testing whether DCCE can successfully identify the individual whale and coda type. Having these tests in place would allow for more growth in the field, potentially enabling real-time identification of calls and individual whales with DCCE.

6. CONCLUSION

DCCE is introduced as the first representation model explicitly designed around the two-channel biological structure of sperm whale codas. DCCE substantially outperforms WhAM on individual identity (F1: 0.834 vs. 0.454, +83% relative) while matching it on social-unit classification (0.878 vs. 0.895), demonstrating that domain knowledge can substitute for data scale on fine-grained identity tasks.

The first layer-wise probing analysis of WhAM was conducted, revealing a recording-year confound (Cramér's $V = 0.51$, $\rho = 0.63$) unreported in the original paper that partially explains WhAM's social-unit advantage. The first controlled synthetic augmentation study for cetacean bioacoustics was also conducted, finding that acoustic plausibility does not guarantee representation benefit when individual-ID labels cannot be assigned to synthetic codas.

The assembled dswp_labels.csv label table and full codebase are released publicly as a resource for future work on the DSWP dataset, enabling reproducible experiments on all four phases.

Beyond the technical results, DCCE has the opportunity to bridge a deeper gap in our understanding of sperm whale communication. Living in tight-knit, multi-generational matrilineal families with communication systems that appear to carry identity, social structure, and potentially culture across generations, sperm whales are amongst the most socially complex creatures. Every coda is not just a click pattern: it is a signal exchanged between individuals who know each other, who have histories, and who depend on acoustic recognition to maintain their bonds across the vast darkness of the deep ocean. DCCE's ability to decode individual and social-unit identity from these signals brings us one step closer to understanding not just how sperm whales communicate, but who is speaking and what they are saying.

7. ACKNOWLEDGMENTS

The authors would like to acknowledge the contributions made by one another throughout this work. Both authors contributed to the exploratory data analysis, paper writing, synthetic augmentation experimentation, and DCCE experimentation and architecture. João led each phase of the project, serving as the primary contributor to the DCCE architecture, including confirming the design and running the final results. Emma tested the DCCE on real-world data, converting YouTube audio to WAV files and evaluating the model with actual sperm whale codas. Emma also contextualized the relationship between coda structure and sperm whale communication. The authors would also like to thank Claude for assistance in organizing the references and QuillBot for grammar and spelling check.

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